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Conveyor belt and modules

Abstract

A modular plastic conveyor belt constructed of injection-molded belt modules having narrowed portions either above or below rod holes in hinge elements. The narrowed portions are formed by a restricted-flow region in an injection-mold cavity that ensures that weld lines in the hinge elements are angled off the pull line of the conveyor belt.

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Background/Summary

BACKGROUND

(1) The invention relates generally to power-driven conveyors and more particularly to modular plastic conveyor belts and belt modules.

(2) Modular plastic conveyor belts are constructed of a series of rows of one or more belt modules linked end to end by hinge rods through laterally aligned passageways of interleaved hinge elements of adjacent rows. The belts are positively driven by drive sprockets or drums along conveying paths that may include straight or curved segments, or both. Some conveying paths with curved segments require the belt to turn in only one direction. Others require the belt to negotiate both left and right turns. In turns the hinge elements closer to the radially outer edge of the belt bear

the belt tension. When a plastic conveyor belt module is injection-molded, molten plastic is injected into a mold's cavity. The flow into the portions of the cavity forming the hinge elements bifurcates around a core pin forming the rod holes. A weld line forms in each hinge element where the bifurcated flows meet. If the hinge elements are designed to be symmetrical about a plane through the rod holes and parallel to the module's pull direction, the weld lines will often be formed between the rod holes and the ends of the hinge elements at or close to the pull line of the module when in a conveyor belt. Belt tension concentrated in the hinge elements around the weld lines often causes the module to break at one or more of the weld lines.

SUMMARY

(3) One version of a conveyor belt module that extends in length from a first end to a second end, in width from a first side to a second side, and in thickness from a top to a bottom comprises an intermediate portion and a plurality of first and second hinge elements. The first hinge elements extend outward in a first length direction from the intermediate portion to a distal end at the first end of the conveyor belt module. Each has a first side wall and a second side wall and a first rod hole extending in a width direction from the first side wall to the second side wall. The second hinge elements extend outward in a second length direction opposite to the first length direction from the intermediate portion to a distal end at the second end of the conveyor belt module. Each has a first side wall and a second side wall and a second rod hole extending in the width direction from the first side wall to the second side wall. The first and second rod holes define a central plane that is parallel to the first and second length directions and to the width direction. The first and second side walls of at least some of the first hinge elements and the second hinge elements each define an upper region between the central plane and the top of the conveyor belt module and a lower region between the central plane and the bottom of the conveyor belt module. The first and second side walls are closer together in a narrowed portion of the lower region than in the upper region or else are closer together in a narrowed portion of the upper region than in the lower region.

(4) Another version of a conveyor belt module that extends in length from a first end to a second end, in width from a first side to a second side, and in thickness from a top to a bottom comprises an intermediate portion and a plurality of first and second hinge elements. The first hinge elements extend outward in a first length direction from the intermediate portion to a distal end at the first end of the conveyor belt module. Each has a first rod hole extending in a width direction through the first hinge element. The second hinge elements extend outward in a second length direction opposite to the first length direction from the intermediate portion to a distal end at the second end of the conveyor belt module. Each has a second rod hole extending in the width direction through the first hinge element. The first and second rod holes define a central plane that is parallel to the first and second length directions and to the width direction and that separates the first and second hinge elements into an upper region above the central plane and a lower region below the central plane. The minimum cross section of the lower region in a plane perpendicular to the first and second length directions in some of the first and second hinge elements is less than the minimum cross section of the upper region, or vice versa.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a bottom plan view of a portion of one version of a belt module for a modular plastic conveyor belt. FIG. 1A is an enlarged axonometric view of a portion of the belt module of FIG. 1.

(2) FIG. 2 is a bottom plan view of a portion of a second version of a belt module for a modular plastic conveyor belt. FIG. 2A is an enlarged axonometric view of a portion of the belt module of FIG. 2.

(3) FIG. 3 is a bottom plan view of a portion of a third version of a belt module for a modular

plastic conveyor belt. FIG. 3A is an enlarged axonometric view of a portion of the belt module of FIG. 3.

(4) FIG. 4 is an enlarged side elevation view of a belt module as in any of FIGS. 1-3.

(5) FIG. 5 is a top plan view of the complete belt module as in any of FIGS. 1-3.

(6) FIG. 6 is a top plan view of a portion of a modular plastic conveyor belt negotiating both left and right turns.

DETAILED DESCRIPTION

(7) One version of a conveyor belt module is shown in FIGS. 1 and 1A from the bottom side and in FIG. 5 from the top side. (FIG. 1 shows only a portion of the complete module of FIG. 5.) The module **10** extends in length from a first end **12** to an opposite second end **13**, in width from a first side **14** to an opposite second side **15**, and in thickness from a top **16** to a bottom **17**. First hinge elements **18**, **18'** extend outward in a first length direction **20** from an intermediate portion **22** of the belt module **10** to distal ends. Second hinge elements **19**, **19'** extend outward from the intermediate portion **22** in an opposite second length direction **21** to distal ends. The first hinge elements **18**, **18'** are offset laterally in the width direction from the second hinge elements **19**, **19'**. Gaps **24** are formed between laterally adjacent hinge elements. First and second rod holes **26**, **27** extend in the width direction through the first and second hinge elements **18**, **18'**, **19**, **19'** from first side walls **28**, **28'** to second side walls **29**, **29'**. A wide edge portion **30** also has rod holes aligned with the rod holes **26** of the first hinge elements **18**, **18'**.

(8) As shown in FIG. 4, the first and second rod holes **26**, **27** define a central plane **32** that is parallel to the first and second length directions **20**, **21** and to the width direction **34**. The central plane **32** lies between the top **16** and bottom **17** of the module **10**. In this example the rod holes **26**, **27** are elongated in the length direction to allow a belt made of such modules to negotiate turns. The central plane **32** divides at least some **18'** of the first hinge elements into an upper region **36** between the central plane **32** and the top **16** of the module **10** and a lower region **37** between the central plane and the bottom **17** of the module. Similarly, the central plane **32** divides at least some **19'** of the second hinge elements into an upper region **38** and a lower region **39**. As shown, the majority of the top **16** of the module **10** is flat, and the bottom **17** is flat below the rod holes **26**, **27**. The flats are joined by an inwardly curving portion of the bottom **17** that forms a concave lateral channel that extends across the width of the module **10** to enable a conveyor belt constructed of the modules to transition around small-diameter nosebars or nose rollers at the discharge ends of conveyors.

(9) As shown in FIGS. 1 and 1A, the side walls **28**, **29** of the first hinge elements **18'** are closer together in a narrowed portion **40** of the lower region **37** than in the upper region **36**. Similarly, the side walls **28'**, **29'** of the second hinge elements **19'** are closer together in a narrowed portion **40'** of the lower region **39** than in the upper region **38**. Because the narrowed portions **40**, **40'** are in the lower regions **37**, **39**, the molten plastic injected into the mold cavity forming the injection-molded belt module **10** flows outward around core pins inserted into the mold to form the rod holes **26**, **27**. The mold surfaces that form the narrowed portions **40**, **40'** bottleneck the flows **47**, **47'**. Thus, the flows **46**, **46'** through the less restricted mold-cavity regions above the core pins advance faster than the flows **47**, **47'** through the lower restricted regions forming the narrowed portions of the hinge elements **18'**, **19'**. Consequently, the flow fronts advancing around the core pins meet and form weld lines **44**, **45** between the rod holes **26**, **27** and the outside of the hinge elements **18'**, **19'** below the central plane **32**. And the outside ends of the weld lines **44**, **45** are positioned well off the pull line of the module **10**, which coincides with the central plane **32**, as shown in FIG. 4. Thus, a belt constructed of such modules **10** with weld lines angled away from the pull line in that way can withstand greater belt tension.

(10) The hinge elements **18**, **19**, which are farther from first side edge **14** of the module **10** than the hinge elements **18'**, **19'**, do not have a narrowed portion because they form a belt's hinge elements that are distant from the outside of a turn and so do not individually have to bear as much belt

tension as the hinge elements **18'**, **19'** at the outside of the turn. And although the narrowed portions are shown only in the lower regions **37**, **39** of the hinge elements **18'**, **19'**, they could alternatively be formed only on the upper regions **36**, **38**. Or the narrowed portions could be on the upper region of some hinge elements and on the lower region of others. In all cases, the restricted regions in the molds cause the formation of weld lines that angle outwardly from the rod holes upwardly or downwardly away from the pull lines.

(11) The first side walls **28**, **29** are shown in the example of FIG. **1** as mirror images of each other. Likewise, the second side walls **28'**, **29'** are shown as mirror images of each other. But the narrowed portions could be formed in other ways as long as the result is a narrowed portion in the lower (or upper) region compared to the upper (or lower) region formed by a slower flow of plastic through a restricted region of the mold. In the version shown in FIGS. **1** and **1A**, the side walls **28**, **29**, **28'**, **29'** curve inward toward each other to a minimum spacing at a waist below the rod holes **26**, **27**, which gives the narrowed portions a varying cross section. Above the rod holes **26**, **27**, the first and second side walls are flat and do not curve inward. At the minimum spacing of the two conveying side walls **28**, **29**, **28'**, **29'**, the cross sections of the narrowed portions **40**, **40'** in a plane **42** perpendicular to the first and second length directions **20**, **21** are a minimum cross section. The cross sections of the hinge elements **18'**, **19'** at their distal ends and at their connections with the intermediate portion **22** are greater than in the narrowed portions for strength.

(12) FIGS. **2** and **2A** depict another version of a belt module that is similar to the belt module **10** of FIGS. **1** and **1A**. The belt module **50** differs from the belt module **10** of FIG. **1** in the shape of the narrowed portions **52**, **52'** of the hinge elements **54**, **54'** closer to the outer side **56** of the module. In this version the side walls in the narrowed portions **52**, **52'** have flat faces **58**, **59** defining the minimum spacing of the side walls at an extended waist. In FIGS. **3** and **3A**, the belt module **60** has curved side walls **62** as in FIG. **1** on the first hinge elements **64** closer to outer side **66** and side walls **63** with flat faces as in FIG. **2** on the second hinge elements **64'**.

(13) The complete module **10** shown in FIG. **5** includes a wider interior hinge element **70** along the module's second end **13**. The underside of the wider hinge element **70** includes a recessed pocket that receives the drive tooth of a drive sprocket to drive a belt constructed of such modules. The pocket also receives the teeth of idle sprockets on a conveyor.

(14) A portion of a conveyor belt **80** constructed of belt modules **10** is shown in FIG. **6** negotiating left and right turns. The belt **80** is constructed of a series of rows **82** of one or more belt modules joined by hinge rods (not shown) extending through lateral passageways formed by aligned rod holes of interleaved first and second hinge elements of adjacent rows **82** at hinge joints **84**. With elongated rod holes as in FIG. **1A**, which allow the belt to collapse at either side **86**, **87**, the belt **80** can make both left and right turns as well as run straight. If the belt is made in a bricklay pattern, each row **82** is constructed of two or more side-by-side modules separated at seams. Alternatively, the belt **80** can be constructed of rows **82** containing a single wide module that defines the width of the belt. The geometry of the belt modules provides no large openings through the belt **10** and so minimizes finger pinching.

(15) As FIGS. **5** and **6** show, the edge portions **30** of the belt modules **10** each include an opening **90** in the top **16** that is shaped to indicate a direction. In this example the shape is a combination of a rectangular portion **92** and a triangular portion **94**. The shape of the opening **90** indicates a direction pointed to by the vertex **96** of the triangular portion **94** that is opposite the rectangular portion **92**. The direction indicated by the openings **90** indicates a direction of belt travel **98** of the conveyor belt **80** constructed of the modules **10**. In other belt versions the openings can be used as an aid in the assembly of the conveyor belt **80** by prompting belt assemblers to install the modules **10** with the openings arranged in the same direction. Other shapes for the openings can be used to indicate a direction to show the direction of belt travel or to aid assembly. Some examples are lone triangles, chevrons, arrows, and arrowheads.

(16) The design of the hinge elements with the narrowed portions provided by the converging side

walls could be altered for different belt applications. For example, the narrowed portions could be formed on some or all the hinge elements with circular, rather than elongated, rod holes for straight-running belts. As another example, the hinge elements with the narrowed portions could be located on only one lateral side of belts designed to turn in only one direction. Or all the hinge elements could be formed with narrowed portions either above or below the rod holes.

Claims

1. A conveyor belt module extending in length from a first end to a second end, in width from a first side to a second side, and in thickness from a top to a bottom, the conveyor belt module comprising: an intermediate portion; a plurality of first hinge elements extending outward in a first length direction from the intermediate portion to a distal end at the first end of the conveyor belt module and each having a first side wall and a second side wall and a first rod hole extending in a width direction from the first side wall to the second side wall; a plurality of second hinge elements extending outward in a second length direction opposite to the first length direction from the intermediate portion to a distal end at the second end of the conveyor belt module and each having a first side wall and a second side wall and a second rod hole extending in the width direction from the first side wall to the second side wall; wherein the first and second rod holes each define a central plane parallel to the first and second length directions and to the width direction; wherein the first and second side walls of at least some of the first hinge elements and the second hinge elements each define an upper region between the central plane and the top of the conveyor belt module and a lower region between the central plane and the bottom of the conveyor belt module; wherein the first and second side walls are closer together in a narrowed portion of the lower region than in the upper region or else are closer together in a narrowed portion of the upper region than in the lower region; wherein weld lines are formed in the upper or lower regions having the narrowed portion.
2. The conveyor belt module as claimed in claim 1 wherein the first and second hinge elements in which the first and second side walls are closer together in the narrowed portion are closer to the first side of the conveyor belt module than are the other first and second hinge elements.
3. The conveyor belt module as claimed in claim 1 wherein the first and second hinge elements in which the first and second side walls are closer together in the narrowed portion are closer to the first and second sides of the conveyor belt module than are the other first and second hinge elements.
4. The conveyor belt module as claimed in claim 1 wherein the first and second side walls are mirror images of each other.
5. The conveyor belt module as claimed in claim 1 wherein the rod holes are elongated in length and wherein the narrowed portion is either above or below the elongated rod hole.
6. The conveyor belt module as claimed in claim 1 wherein the first and second side walls in the narrowed portion of the first hinge elements are curved and wherein the first and second side walls of the second hinge elements have flat faces in the narrowed portion.
7. The conveyor belt module as claimed in claim 1 wherein the narrowed portion of the first hinge elements is in the upper region and the narrowed portion of the second hinge elements is in the lower region.
8. A conveyor belt comprising: a series of rows of one or more conveyor belt modules including conveyor belt modules as claimed in claim 1; a plurality of hinge rods extending through lateral passageways formed by the aligned first and second rods holes of the interleaved first and second hinge elements of the one or more conveyor belt modules of adjacent rows to join the rows together at hinge joints between adjacent rows.
9. A conveyor belt module extending in length from a first end to a second end, in width from a first side to a second side, and in thickness from a top to a bottom, the conveyor belt module

comprising: an intermediate portion; a plurality of first hinge elements extending outward in a first length direction from the intermediate portion to a distal end at the first end of the conveyor belt module and each having a first rod hole extending in a width direction through the first hinge element; a plurality of second hinge elements extending outward in a second length direction opposite to the first length direction from the intermediate portion to a distal end at the second end of the conveyor belt module and each having a second rod hole extending in the width direction through the second hinge element; wherein the first and second rod holes each define a central plane parallel to the first and second length directions and to the width direction and separating the first and second hinge elements into an upper region above the central plane and a lower region below the central plane; wherein the minimum cross section of the lower region in a plane perpendicular to the first and second length directions in some of the first and second hinge elements is less than the minimum cross section of the upper region, or vice versa; wherein weld lines are formed in the upper or lower region having the lesser minimum cross section.

10. The conveyor belt module as claimed in claim 9 wherein the first and second rod holes are elongated in length and wherein the lesser minimum cross sections are either above or below the elongated first and second rod holes.

11. The conveyor belt module as claimed in claim 9 wherein the first and second hinge elements having the lesser minimum cross section in either the upper or lower region are closer to the first and second sides of the conveyor belt module than are the other first and second hinge elements.

12. The conveyor belt module as claimed in claim 9 wherein the first and second hinge elements have first and second side walls and wherein the first and second side walls in either the upper region or the lower region curve toward each other from opposite ends to a waist defining a minimum spacing of the first and second side walls.

13. The conveyor belt module as claimed in claim 12 wherein the first and second side walls are mirror images of each other.

14. A conveyor belt module comprising: a first end and an opposite second end defining the length of the conveyor belt module; a first side and an opposite second side defining the width of the conveyor belt module; a top and a bottom defining the thickness of the conveyor belt module; an outer edge portion at the first side including an opening having a shape that indicates a direction.

15. The conveyor belt module as claimed in claim 14 wherein the opening opens onto the top of the conveyor belt module.

16. The conveyor belt module as claimed in claim 14 wherein the opening has a shape having a rectangular portion and a triangular portion pointing in the direction.

17. The conveyor belt module as claimed in claim 14 wherein the direction indicated by the opening indicates a direction of belt travel of a conveyor belt in which the conveyor belt module is installed.

18. A conveyor belt constructed of conveyor belt modules as claimed in claim 14 such that the openings of all the conveyor belt modules when installed in the conveyor belt indicate the same direction.
